Local NS-NS merger rate density [Gpc^{-3} yr^{-1}]
10^{-3} 10^{-2} 10^{-1} 10^{0} 10^{1} 10^{2} 10^{3} 10^{4} 10^{5}
GWTC-2 (Abbott et al. 2020)
GWTC-3 (Abbott et al. 2021)
GWTC-4 (Abbott et al. 2025)
GWTC-5 (Abbott et al. 2025)
Gravitational waves
Coward et al. (2012)
Petrillo et al. (2013)
Fong et al. (2015)
Della Valle et al. (2018)
Jin et al. (2018)
Zhang et al. (2018)
Dichiara et al. (2020)
Zevin et al. (2022)
Short gamma-ray bursts
Jin et al. (2016)
Doctor et al. (2017), DES
Kasliwal et al. (2017), PTF
Smartt et al. (2017), ATLAS
Yang et al. (2017), DLT40
Andreoni et al. (2021), ZTF
Kilonovae
O'Shaughnessy and Kim (2009)
Kim et al. (2015)
Pol et al. (2020)
Grunthal et al. (2021)
Galactic pulsar binaries
O'Shaughnessy et al. (2010)
Lipunov and Pruzhinskaya (2014)
Mennekens and Vanbeveren (2014)
Dominik et al. (2015)
de Mink and Belczynski (2015)
Ablimit and Maeda (2018)
Belczynski et al. (2018a)
Chruslinska et al. (2018)
Giacobbo and Mapelli (2018)
Giacobbo and Mapelli (2020)
Klencki et al. (2018)
Kruckow et al. (2018)
Mapelli and Giacobbo (2018)
Vigna-Gómez et al. (2018)
Artale et al. (2019)
Baibhav et al. (2019)
Boco et al. (2019)
Chruslinska et al. (2019)
Eldridge et al. (2019)
Neijssel et al. (2019)
Belczynski et al. (2020)
Santoliquido et al. (2020)
Tang et al. (2020)
Zevin et al. (2020)
Chu et al. (2021)
Ghodla et al. (2021)
Olejak et al. (2021)
Santoliquido et al. (2021)
Briel et al. (2022b)
Broekgaarden et al. (2021, 2022)
Olejak et al. (2022)
Romagnolo et al. (2023)
Boesky et al. (2024)
Deng et al. (2024)
de Sá et al. (2024)
Chattaraj (2025; solar metallicity)
Chen et al. (2025)
Marinacci et al. (2026)
Mestichelli et al. (2025)
Pellouin et al. (2025)
Romagnolo et al. (2025)
De Santis et al. (2026)
Sgalletta et al. (2026)
Isolated binary evolution
Hamers and Thompson (2019)
Vynatheya and Hamers (2021)
Triples / Multiples
Lee et al. (2010)
Bae et al. (2014)
Samsing et al. (2014)
Belczynski et al. (2018a)
Ye et al. (2020)
Globular clusters
Antonini and Perets (2012)
Petrovich and Antonini (2017)
Belczynski et al. (2018a)
Fragione et al. (2019)
McKernan et al. (2020)
Wang et al. (2021)
Nuclear star clusters
Ziosi et al. (2014)
Fragione and Banerjee (2020)
Santoliquido et al. (2020)
Young / Open star clusters
10^{-3} 10^{-2} 10^{-1} 10^{0} 10^{1} 10^{2} 10^{3} 10^{4} 10^{5}
Local NS-NS merger rate density [Gpc^{-3} yr^{-1}]